



Blacktown
City Council

**BLACKTOWN LEISURE CENTRE STANHOPE
MAIN POOL HALL HEATING & VENTILATION
AHU 1 & 3 REFURBISHMENTS**

TECHNICAL REQUIREMENTS

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1. GENERAL REQUIREMENTS

1.1 General

The work covered under this Specification will generally replicate the works previously carried out on AHU 2 including identical performances and sizes (we encourage the tenderers to review the newly refurbished AHU 2) and comprises the following:

1. Removal and disposal of two (2) existing belt driven exhaust fans.
2. Removal and disposal of the two (2) sets of air-to-air heat exchangers
3. Removal and disposal of the exhaust plenum floors and replace with non-skid aluminium plates
4. Supply and install two (2) sets of new run-around heat recovery coils (4 in total) and dedicated water recirculation pumps (2 in total) and respective VSDs.
5. Supply and install four (4) new evaporator coils and corresponding two (2) condenser coils.
6. Supply and install two (2) new water-cooled heat pump chiller with inbuilt sets of chilled and condenser water circulation pumps, including 2x 90mm high concrete plinth to sit each chiller respectively. Pouring of the concrete on the plinth frame shall be carried out ahead of time allowing complete curing before the chillers' delivery.
7. Supply and install drift eliminators at the upstream of the evaporator coils similar to existing.
8. Supply and install condensate trays common to the evaporator coils and the cool side of the heat recovery coils.
9. Supply and install eight (8) EC plug-in exhaust fans and accessories.
10. Replace the front and top panels of the AHUs to improve its structural strength, material similar to AHU-2.
11. Supply and install new lockable access doors, material similar to AHU-2, and re-use the other respective access doors.
12. Modify and make good AHU 1 & 3 internal unit panels and apply rust protection wherever necessary.
13. Modify exhaust ductworks to integrate with the new AHUs discharge reconfiguration.
14. New interconnecting heat recovery pipework between the run-around coils of AHU 1 & 3 and pumps.
15. New interconnecting CHW and COW pipework between AHU 1 & 3 and the new Chillers.
16. Associated electrical and interconnecting controls work.
17. Modify and reconfigure the existing electrical system to accommodate the chillers and other associated equipment.
18. Associated plumbing/hydraulics works.
19. Associated builder's work, making good, removal of redundant items and construction rubbish, etc.
20. All other works not mentioned above to form a complete and operating system similar to the newly refurbished AHU 2.

The plantroom external wall louvres shall be removed temporarily to allow removal of the larger redundant items in the plantroom and delivery of the new items into the plantroom including the chillers, coils etc, all in one operation to be agreed and arranged with BVM/BLCS management. Re-

instate the wall louvre as soon as delivery, hauling and crantage are completed and make-good any disturbed items affected when the louvre was removed and during the delivery process.

The work is to be done in a coordinated/programmed manner to suit the operational requirements of the centre and with AHU 2 to operate continuously except during the mechanical switchboard upgrade.

All materials and workmanship shall comply with the relevant Australian Standard and Regulatory Authority requirements.

1.2 Licensed Personnel

All work shall be carried out and supervised by suitably qualified and certified/licensed tradespersons, experienced in the relevant trade.

All aspects of the installation will be installed and constructed to comply with WorkSafe NSW regulations and Council WHS requirements.

1.3 Drawings and Sketches

The following drawings are issued for information purpose:

M114 - Mechanical Services – Air & Water Schematic - Existing

Existing Equipment Schedule

SKM-01 – AHU -1&3 Piping Schematic Drawing

SKM-02 – Existing Power Supply Diagram

SKM-03 – Proposed Power Supply Diagram

SKM-04 – Plantroom Layout

SMK 05 - EC Fans External Wiring Diagram

1.4 Programming of the Work

Due to the nature of the work it is brought to the Contractors' attention that programming of the work and subsequent site management of the work are considered essential to minimise disruption to the leisure centre's activities.

Immediately following award, the Contractor shall submit to Council a detailed schedule showing the proposed sequence of work and the durations for which the various areas will need to be available to the Contractor for the work to be completed. This shall be based on the schedule and time for completion submitted with the Tender.

1.5 Centre's Opening Hours

Leisure centre's hours are 6:00am to 9:00pm daily. Any noisy works shall consider the neighbouring properties such as any noisy works shall be limited to typical working hours or agreed with the leisure centre's management.

1.6 Interruption of Existing Services

A shutdown, where necessary, shall be arranged at a time to suit the leisure centre's management. At least 48-hour notice in writing of electrical shutdown must be given and approval obtained in writing. Such shutdown time shall be kept to a minimum. Tenderers shall allow accordingly in their Tender. No claims for extra cost in this respect will be considered.

1.7 Dirt, Dust, Noise and Vibration Control

During the progress of the works, the leisure centre will remain occupied and operational.

Take precautions to protect the existing building, areas that are adjacent to any work location, and their contents. Minimise dirt, dust and nuisance and collect and remove all rubbish and debris on a daily basis. Where work is carried out in occupied or partially occupied areas the Contractor shall

ensure that the work is carried out with a minimum of nuisance or annoyance to the centre's staffs and patrons.

Explosive powered tools are not permitted.

Transmission of excessive vibration to the building structure will not be allowed at any time. If directed by a centre's staffs, any work which they consider is causing, or is likely to cause, operational problems, shall be ceased immediately until an alternative method is agreed.

Drop sheets and covers shall be provided over all plant and equipment liable to damage from dirt, dust and debris. When working over the top of equipment, utmost care must be taken to protect the equipment from mechanical damage, heat damage, from dust and debris falling as a result of drilling and fixing activities, and from painting.

The leisure centre staffs are currently using some plantroom space for storage of their consumables. Contractor to take consideration of the safety of the leisure centre staffs and provide safe passage through the work area. If necessary, assist in relocating some of the shelves to allow safe access to their consumables.

1.8 Deliveries to Site and Off Loading

When dispatching plant or materials to site, or removal of plant or materials from site, the Contractor shall arrange to have a responsible person on site to supervise and arrange the offloading/loading. The Contractor shall supply his own labour and equipment necessary for the offloading and hoisting into position of plant and materials, and shall arrange for the storage and safekeeping of such plant and materials. Plant and materials shall be dispatched to site to suit the installation schedule and minimise double handling on site.

1.9 Hoisting & Cranage

Provide all cranage, lifting, hauling, hoisting, protection, hoarding, scaffolding (if applicable) and the like associated with the works of the Contract to the requirements of SafeWork NSW, Service Australia, Blacktown Council and any other relevant authority having jurisdiction. Pay all fees associated therewith and allow accordingly in the Tender.

It shall be the Contractor's responsibility to propose the Cranage Strategy and carry it out to the satisfaction of the Council.

Ensure that all equipment selected shall fit in the hoisting facilities available before ordering. Under no circumstances is any hoisting of equipment to be carried out unless written approvals have been given by Council.

The Contractor shall ensure that any and all works are undertaken fully in accordance with proper BCC WHS requirements.

The Contractor shall be responsible for all hoisting and cranage associated with the Work and shall allow for the number of cranage lifts to suit the staged nature of carrying out the work.

1.10 Making Good

All disturbed work, all damage that may occur and all surfaces and penetrations affected as a result of the works, including damage to the grounds during crane lift, covered by this Specification shall be made good to the entire satisfaction of Council.

On completion of the works the Contractor shall remove all debris, clean-up site and leave the site in a tidy condition.

1.11 Recycling of Redundant Equipment

The Contractor shall ensure that all materials removed from the site, and which can be recycled, shall be transported to a recycling depot. This includes items such as existing steel, copper, pipework, wiring, etc. Plastic components that cannot be recycled shall be disposed of in accordance with statutory requirements.

The Contractor shall submit evidence that removed material has been transported to a recycling depot.

2. DESCRIPTION OF WORKS

2.1 Introduction

This project aims to enhance pool hall space heating and ventilation by introducing energy-efficient solutions. Electric-powered heat pump chillers will replace natural gas heating, significantly reducing gas reliance. Water-cooled chillers for AHU-1 and AHU-3 will supply heating hot water, offering triple the energy efficiency of conventional gas-fired boilers. Corroded exhaust fans will be replaced with EC motor-driven fans, achieving approximately 40% fan power savings.

These upgrades of AHU-1 and AHU-3 will align with the proven design of AHU-2 upgraded two years ago, ensuring consistent performance and substantial energy savings while addressing the deteriorated pool hall ventilation system.

2.2 Scope of Works

AHU 1 & 3 Refurbishment and New Heat Pump Chillers for Pool Hall Ventilation and Heating.

2.2.1 Each AHU Plenums & Associated Works

Modify the supply and exhaust chambers to suit new equipment to be installed and as follows:

- a. Remove and dispose existing filter bags. Frames and tracks may be re-used.
- b. Re-instate filter frames into the new outside air intake centre compartment to fit 8-off 600x600 bag air filters, and then, supply and install new 600x600, 600mm deep bag air filters.
- c. Remove and dispose off-site the following items within the section of AHU 1 & 3 to give way for the repair and modification works:
 - 1) Air-to-air heat exchanger, frames, condensate sump and drainpipes
 - 2) Badly corroded exhaust air fan, discharge duct up to the joint below roof and electricals
 - 3) Front external panels and filter frames, including filter frames on plenum roofs
 - 4) All redundant items and other items not required in the proposed modification.
- d. Sand to remove (minor) rust, rust treat surface and coat with paint as necessary
- e. Install new minimum 2.0mm or thicker aluminium checkered plates over the existing discharge plenum floors.
- f. Remove section of sheet metal ductwork above AHUs to allow connection of the new exhaust discharge configuration.
- g. Supply and install internally-insulated galvanised sheet metal ductwork above AHUs to connect new exhaust plenum to existing exhaust ductwork
- h. Re-enforce existing galvanised sheet metal partition between AHUs to withstand exhaust air pressure
- i. Patch up gaps with galvanised sheet metal between coils and walls to minimise air leakages
- j. Supply and install new large lockable access panel for service access to each AHUs
- k. Clean, repaint and re-instate existing AHU access doors. Council to install safety warnings on the access doors.
- l. All physical dimensions and suitability to be confirmed on site before manufacture.

2.2.2 Mechanical & Associated Works

- a. Design, supply and install condenser and evaporator water coils including insulated pipework, corrosion-resistant sump, drain pipes, supports, fittings, etc. Chlorine resistant coating shall be applied to coil fins.
- b. Design, supply and install heat recovery coils including recirculating pumps, pipework, corrosion-resistant sump, drain pipes, supports, fittings, etc. Chlorine resistant coating shall be applied to coil fins.
- c. Supply, deliver and install new EC fans (4-off) in each AHUs at 2,800 l/s capacity each to replace the badly-corroded centrifugal exhaust fans, including building up of adequate fan supports and re-enforced exhaust plenums
- d. Design, supply, deliver and install finned coils in the fresh air and exhaust air plenums sized based on the proposed heat pump chiller performance
- e. Supply and install aluminium drift eliminators upstream of the evaporator coils
- f. Supply and install condensate sumps common to the cool side of the heat recovery coils, evaporator coils including the drift eliminators.
- g. Interconnecting pipeworks into each system including valves, fittings and other essential items associated works such as bleeds, drains, etc
- h. Supply and install all associated and interconnecting pipework, make-up water/RPZ valve, chemical dosing, closed-type expansion tank valves, fittings, spring/rubber isolation, gages cocks, installation of thermowells (supplied by others), etc to form a complete and operational chilled/condenser water system. All pipework and electrical cabling/trunking to run 2.1m min from floor in trafficable area.
- i. Supply and install 80mm table B copper chilled and condenser water pipework with necessary valves, brackets and fittings from new headers to connect 3x coils in each AHUs
- j. Fill and pressure test pipe works using minimum 3x the system pressure.
- k. Insulate chilled water pipework using foil faced foam shield type pipe insulation and metal sheath at low level pipework (under 1.8m).

2.2.3 Upgrade of Building Management Control System & Associated Works (by Others)

Note: HVAC and BMCS contractors shall be responsible for coordinating inter-related works and no waiver of responsibility is allowed. Advise the project officer any conflict between the scope and responsibility.

- a. Extend the existing BMCS to control the new works, logic shall be identical to the newly refurbished AHU 2 control
- b. Supply and install necessary and additional points to control and monitor the new equipment
- c. Re-engineer and reprogram BMCS and other necessary works to completely control the new ventilation and heating systems
- d. Integrate the new Chiller BACnet High Level Interface gateway with the BMS to operate as specified and provide the Device Instance and IP details and shall carry out all necessary programming to ensure the current Chiller Staging strategy is maintained.
- e. Test and commission the new installation.
- f. Allow progressive fine-tuning of control logic after Commissioning up to the end of the Defects Liability Period, without any additional cost to the Contract.

2.2.4 Chiller installation & associated works

- a. Clear the area where the proposed heat pump chiller will be positioned, and then, form and pour 90mm high concrete plinths to suit each new chiller. Pour concrete ahead of time to allow the concrete to fully cured before the chiller delivery.

- b. Supply, deliver and install 2-off reverse-cycle packaged heat pump including in-built evaporator and condenser pumps. Chillers shall be Carrier 61WG190A water cooled chiller with heating nominated capacity shall be 212kW or equal. Chillers to include BACnet HLI card.
- c. Supply, deliver and install 2x 61WG190A water cooled chiller into plant room and locate on new concrete plinths
- a. Test and commission chiller to its operating parameters.

2.2.5 Water Treatment

Water Treatment shall be to the Water Treatment Design Guide requirements. Provide separate water treatment to each system including dosing pot on a SS Tray under the feet and drained to waste for each closed system.

Water quality shall be tested and chemical treatment concentration to be adjusted every three months.

2.2.6 Electrical & Associated Works

- a. Upgrade or replace the old non-compliant MCC1 in accordance with the relevant and current electrical code including replacement of electrical components within the board if necessary.
- b. Disconnect power supply cables from the main switchboard into the MSSB-1 and reconnect the power supply cables into the new MSSB-2(old MCC1).
- c. Supply and install adequately sized power cables from the chiller sub-board located in the Main Switch room into existing MSSB-1 located in Plantroom 1.
- d. Supply and install switchgears in MSSB-1 to accommodate new chillers and provide power to the new sub-board, including terminals and relays for BMS interfacing, ensuring correct control and status on the Control.
- e. Supply and install new electrical sub-board, cables and circuit breakers for the auxiliary equipment including heat recovery circulation pumps and EC fans as shown on the drawings.
- f. Supply and install cable tray, cable supports, cabling, isolator and connection where required
- g. Supply and installation of cable support, cabling, isolator and connection to (8) Exhaust Fans in the AHUs and the new chillers.
- h. Supply and install 7.5kW ABB ACH580 VSD's to respective circulating pumps for the heat recovery coils.
- i. All other associated and necessary electrical works not specifically mentioned above complying to current and applicable electrical codes and safe operation of the systems.

2.2.7 General

- a. Remove and re-instate plantroom external wall louvre for access to equipment delivery and retrieval of redundant equipment
- b. Remove any job generated rubbish and dispose off-site, including redundant equipment
- c. Co-ordinate with centre management and Council officer in charge of the project, including allowing safe access to centre's miscellaneous stocks they need for centre's operation and safe movement of the pool staffs through the work area
- d. All other associated works, including controls, builder's, electrical, hydraulics and other associated works not specifically mentioned in this scope of works to form a complete and operating system
- e. Test and commission the system

- f. All works to be performed in good engineering and industry practice, and in accordance with relevant codes, standards and local authority
- g. Provide as installed drawings
- h. Provide maintenance manuals
- i. Provide 12 months defect liability warranty parts and labour on all new work
- j. Provide 12 months maintenance based on 3 x visits and one annual service
- k. Submit three (3) sets of the following:
 - 1) Workshop Drawings; 1:50 Plan, Sections and Details.
 - 2) Electrical Single Line Diagram.
 - 3) Automatic Controls Schematic Diagrams
 - 4) "As-Installed" Drawings including one (1) Autocad and PDF USB

3. EQUIPMENT

3.1 New Reverse Cycle Water-Cooled Heat Pump Chiller

3.1.1 General

The chiller shall be factory assembled and tested, ready to operate prior to delivery to Site, with all interconnecting refrigerant piping and wiring ready for field installation. The chiller shall be pressure tested, evacuated and fully factory charged with refrigerant. The chiller shall be factory tested at its full thermal load and each capacity step.

The chiller shall consist of scroll type compressors with soft start capability, condenser, evaporator, microprocessor controls and user display. The chiller shall be complete with a factory installed main circuit breaker with electrical disconnect/isolation capability.

The chiller shall incorporate in-built chilled and condenser water pumps sized to handle the chiller's total nominated water flow at least 400kPA capacity driven by a VSD and controlled by the chiller's internal controller.

Service availability, technical support and spare parts shall be readily available in Australia. The manufacturer's recommendations shall be followed in regards to handling, installation, commissioning and ongoing maintenance. The chiller manufacturer's local distributor shall be engaged by the Contractor for maintenance during the Defects Liability Period.

3.1.2 Product Data

Provide the following information with the Tender:

- 1. Manufacturer's rated performance data for the Chiller offered.
- 2. Full dimensional details in the form of certified drawings.
- 3. Weights, including loading diagrams.
- 4. Details of facilities and services required for the Chiller offered.
- 5. Details of accessories and features provided with the Chiller offered.
- 6. Electrical requirements for the Chiller offered.
- 7. Wiring Diagrams.
- 8. Evidence of conformance with AS1677.2 and AS1210.

3.1.3 Accessories

The following shall be provided with the Chiller or as part of the pipework modifications to connect the new Chiller into the CHW piping reticulation:

1. Water flow and return connections. Flanged to AS2129 Table E or mechanical rolled-groove couplings.
2. Shipping protection; fit blank companion flanges, bolts and gaskets or removable caps.
3. Drain connections; minimum 20dia. Screwed connection with valve at lowest point of evaporator.
4. Vent connections; minimum 15dia. Screwed connection with valve.
5. Thermometer pockets; provide at water inlet and outlet.
6. Water side pressure drop test sockets.

3.1.4 Unit Construction

The chiller frame shall be of heavy gauge, painted galvanized steel. Side enclosure panels shall be provided to provide protection to the condenser coil and piping. Manufacturer's standard finish shall be acceptable and shall be not less than oven-baked powder coating.

The chiller shall be complete with seismic restraint spring supports selected by the chiller manufacturer and suitable for the operating weight of the Chiller. Minimum vibration isolation efficiency shall be 95%.

3.1.5 Compressors

Compressors shall be of the full-hermetic scroll type with integral oil level sight glass. Each compressor shall be driven by a 2-pole electric motor cooled by suction gas and protected by internal temperature sensors. Over temperature motor protection shall also be provided.

Each compressor shall be fitted with "soft start" control complete with phase loss protection. Each compressor shall be fitted with an electric crankcase heater.

Each compressor assembly shall be anti-vibration mounted to the chiller frame.

Each refrigerant circuit shall be complete with the following:

1. Suction and discharge line isolation valves.
2. Replaceable core filter drier.
3. Moisture indicating sight glass.
4. Electronic expansion device.
5. Liquid line service valves.

3.1.6 Evaporator Heat Exchanger

The evaporator shall be a plate heat exchanger type fabricated from Type 316L stainless steel, copper brazed. Evaporators shall be overall insulated with 19mm closed cell foam type insulation. Evaporator shall be fitted with a factory installed electronic flow switch.

Evaporator shall be tested and stamped in accordance with European PED 97/23/EC Code.

3.1.7 Condenser Heat Exchanger

The condenser shall be a plate heat exchanger type fabricated from Type 316L stainless steel, copper brazed. Condenser shall be overall insulated with 19mm closed cell foam type insulation. Evaporator shall be fitted with a factory installed electronic flow switch.

Condenser shall be tested and stamped in accordance with European PED 97/23/EC Code.

3.1.8 Evaporator and Condenser Pumps

Evaporator and condenser circulating pumps shall be in-built with recommended matching capacities with the respective chiller's heat exchangers.

The pumps shall be speed controlled by a variable speed drive control primarily to improve part load performance, reduce noise level and to increase service life.

Enclosures for the variable speed drives shall have a degree of protection of IP55.

3.1.9 Controls

The man-machine-interface shall be via a 180mm (nominal 7") touch screen, finger or touch-pencil capable. The control system shall be similar to Carrier "Touch Pilot" with the "Energy Management Module", or approved equivalent.

The touch screen shall permit web-connectivity, BACnet IP connectivity, trending capability and downloadable documentation capability.

BMS Interface shall be factory installed to provide a high-level interface (HLI - BACnet over IP Gateway) with the existing BMS. This shall enable the BMS system to interrogate the control panel, reset the chiller setpoints, and manage alarms. Allow for necessary field programming.

Set the Device Instance and IP details and provide a mapping table for the HLI points.

The following safeties shall be provided:

1. Reverse rotation (phase reversal).
2. Low Water Temperature.
3. Low Water Flow Rate.
4. Motor Thermal Overload.
5. High Pressure Cut-out.
6. Low Suction Pressure.
7. Limit Start Timer.
8. Electrical overload.
9. Loss of Phase.
10. Low Voltage Power Supply Failure.

Provide a 240V/1Ph/50Hz power supply with GPO for connection of a laptop computer.

3.2 Coil Design and Performance

3.2.1 General

Coil configurations shall be designed to allow for condensate to flow away from the coil easily without being drawn through the coil into the air stream or waterlogging the coil.

Coils shall be copper tube with aluminium fins, copper headers and aluminium casing. Tube lengths and fins shall be free from joints. Coils shall be arranged in slide racks for ease of removal. No non-ferrous parts shall be in contact with any ferrous part.

Each AHU must be equipped with a single coil capable of meeting the specified heating and cooling capacities in both heating and cooling modes.

All coils shall have ARI 410 certification for performance verification.

3.2.2 Condensate trays

Provide condensate tray or trays to collect all condensation occurring inside the unit, including from the drift eliminators. A trapped drain to waste shall be provided to maintain a sufficient column of water to provide an airtight seal under the greatest suction through pressure created by the unit. The drain shall terminate above a tundish or floor waste that is readily accessible and sighted for maintenance.

Material: Aluminium to AS/NZS 1734, or stainless-steel sheet grade 304L.

3.3 EC Plug Fans

3.3.1 Construction

Fan Casing:

The casing shall be painted hot dip galvanised mild steel or aluminium alloy of welded and bolted construction. Include arrow showing direction of rotation.

Drive:

Fan drive to be either IEC direct drive motor or plug fan manufacturer's variable speed integrated motor with on board replaceable EC or VFD electronics.

Impeller:

Hot dip galvanised mild steel, balanced and designed for continuous operation.

Guarding:

Guarding to AS shall be incorporated and applied to the internal face of the air-handling unit to prevent accidental contact with the impeller on opening of AHU inspection doors. Guarding over or around the impeller is not recommended as it imposes additional losses.

4. PIPEWORK

4.1 General

All piping before delivery to site and pending erection shall be sealed against dirt and moisture. Before erection, pipes and fittings shall be cleaned of scale, burrs or other obstructions.

Welded or brazed joints shall be made by experienced and fully qualified tradesmen and in accordance with best trade practice. Dry nitrogen shall be used in all instances.

Each part of the system of piping shall be complete in all detail and provided with all control valves and fittings necessary for satisfactory operation.

The piping shall be accessible for repairs and provided with sufficient flanged and/or flared fitting joints for disconnecting and removing equipment.

Where piping is exposed and liable to damage, it shall be suitably protected with well braced 1.6mm zinc anneal sheet metal covers or heavier section if necessary.

4.2 Pipe Materials

New piping shall be copper solid drawn to AS1432 Type B and shall be installed in accordance with AS1135 (Non-Ferrous Piping Code). All joints in copper pipe shall be silver soldered with silver content of not less than 5%. All silver soldering shall be carried out by skilled tradesmen having recognized training and experience in this type of work. Adequate fire precautions shall be taken as per AS1674. Pipe lengths and materials shall be stored at site with end caps, to prevent ingress of dirt and debris into the piping.

4.3 Pipe Supports

All pipes must be adequately and substantially supported and restrained both horizontally and vertically using a proprietary support system. Pipework adjacent to equipment mounted on vibration isolation mounts must be arranged to provide adequate flexibility to ensure vibration is not transmitted to the building structure. Sprung pipe supports are required to prevent damage to the pipes.

All supports must be constructed from zinc plated galvanised steel with contact between dissimilar metals prevented by non-conducting isolating materials. A specially moulded PVC insulation strip will be used as supplied by the clamp supplier. Tape will not be used in any circumstances.

Support type: Proprietary metallic-coated steel channel section with clamps and hangers sized match external diameter of pipe being supported. Fit moulded plastic end caps to all channels.

Vertical pipes: Provide anchors and guides to maintain long pipes in position and supports to balance the mass of the pipe and its contents.

Change in Direction: Provide a support at each change of direction.

Saddles: Do not provide saddle type supports for pipes $\geq$ DN 25.

Pipe support spacing table

Nominal Pipe Size, DN	Maximum spacing (m)	
	Horizontal	Vertical
25	2	3
32	2.5	3
40	2.5	4
50, 65, 80 & 100	3	4

4.4 Pipe Identification

Pipework and valves shall be clearly labelled and identified to AS 1345-1995. Lettering on piping 50 mm in diameter, and smaller, shall be 13 mm high, and all lettering on larger diameter piping shall be 20 mm high. Direction of flow shall be clearly marked below each pipe label.

4.5 Pipe Insulation and Cladding

All Chilled Water pipework and fittings shall be insulated. Insulation shall be of FM Global approved material, of a thickness to meet the requirements of the Building Code of Australia.

Prior to installation of the insulation, metal surfaces shall be clean and dry and ensure the insulation is dry. Insulation shall be held tight to the pipe using 3 loops of galvanised or stainless-steel wire per insulation section, with the edges and ends tightly butted together. At pipe support locations the insulation shall be tightly butted up to the spacer blocks and sealed using aluminium adhesive tape. All insulation and facing shall be applied strictly in accordance with the manufacturer's recommendations.

Chilled and Condenser Water pipework 1.8m from floor and lower shall be sheathed with 0.6mm zinc anneal sheet with powder coat finish to be advised. Joints in sheathing shall be lapped at least 30mm and riveted or screwed at 150mm maximum centres. Neatly cone down the sheathing to flanges and at terminations return the edges of the sheathing to protect the insulation. Provide sheet metal sleeves over the wooden ferrule to protect from soaking from the rainwater.

4.6 Testing of Pipework

Notification of tests shall be given; Council reserves the right to witness the tests. All test results shall be recorded and submitted for approval.

Piping and joints shall be visually inspected for leaks. Defective sections shall be removed and replaced, and the system retested.

Drain and overflow piping shall be water-tested for leaks and for unobstructed drainage.

5. VALVES

5.1 General

Valves should be sized equal to the nominal pipe size, unless a smaller size is required for throttling, balancing and measurement, or for control valve authority.

Insulated valves:

Provide extended shafts or bodies to butterfly and ball valves to allow full thickness of insulation without restricting movement of hand-wheel or lever.

Connections:

- Valves $\leq$ DN 50: Screwed to AS ISO 7.1.
- Valves $>$ DN 50 and valves in headers: Flanged to AS 2129

All valves shall be of globe type, key operated c/w sealed caps.

Where valves do not have direct capillary joint to the pipe, care shall be taken to prevent distortion of the valve or power element by heat.

Valves in pipelines smaller than 25 mm shall be supported independently of the line.

6. DUCTWORK

6.1 General

Construction and installation of ductwork shall comply with AS 4254 – “Ductwork for Air-handling Systems in Buildings”.

Unless specified otherwise (for special systems) ductwork material shall be prime quality lock forming galvanised steel, Grade G2 or G3 to AS 2338 with Z275 coating to AS 1397.

All sheet metal ductwork shall be manufactured from lock forming quality Lysaght Galvabond sheet steel or equal. Galvanising shall be: 380gms/m².

Drive slide cross joints shall be sealed airtight at corners using metal tongues and Emerduct sealer, or similar.

Unless specified otherwise, ducts shall be sealed in accordance with Table 2.2.1 of AS 4254. Drive slide cross joints shall be sealed airtight at corners using metal tongues and Emerduct sealer, or similar.

Leakage rates of unsealed ducts shall not exceed the rates indicated in Table 2.2.2 of AS 4254. Ductwork systems at Pressure Class 750 and less shall be leak tested using pressure testing method.

All obvious leaks shall be sealed with appropriate and approved sealant.

6.2 Duct Thermal Movement and Vibration

All ducts shall be carefully designed and provided with all necessary anchoring and flexible connections to prevent damage to either the ducts or the building structure due to thermal expansion and/or contraction of the ducts and to obviate transmission or vibration from any motive or noise generating equipment such as fans, air handling units and mixing boxes.

All anchor points and flexible connections shall be shown on the shop drawings and specific approval shall be obtained from appropriate authority before proceeding with the installation of any anchors or fixings to the building structure.

6.3 Cleaning and Protection

All ductwork shall be fabricated under cover, delivered to site and stored in a weatherproof and dry area. Open ends of duct sections shall be covered with plastic sheet or tarpaulins until required for installation. Special care shall be given to internally insulated ducts, silencers etc. for protection against dust and moisture.

Ductwork shall not be installed unless adequate cover and protection is available to protect it from possible construction damage and the elements.

Prior to and during installation, ducts shall be thoroughly cleaned out and shall have all ends covered in an approved manner to prevent ingress of dust and general building debris.

6.4 Sealing and Flashing

All ducts protruding through walls and floors and exposed to view shall be provided with neat, galvanised steel sealing frames fixed to ducts and walls or floors as appropriate.

All external installations and roof penetrations shall conform to Section 3 of AS 4254.

6.5 Insulation

All ductwork shall be adequately insulated to prevent thermal losses and condensation forming on the outside of the ductwork and/or insulation and comply with all requirements of Section J of the NCC.

All materials used in the installation of insulation shall have the following resistance properties when tested to AS 1530 Part 3 "Test for Early Fire Hazard Properties of Materials".

Ignitability Index	0
Spread of Flame Index	0
Heat Evolved Index	0
Smoke Developed Index	2

Where air mixing and filter plenums are accessible for filter changing, the plenum floor insulation shall be covered with 1.2 mm thick galvanised steel sheet sufficiently reinforced and supported for rigid construction. The plenum walls and ceiling insulation shall be faced with perforated galvanised steel sheet.

The insulation materials shall be resistant to bacteria, algae, vermin, and mould or fungi growth and be non-hygroscopic.

7. AUTOMATIC CONTROLS (By Others)

7.1 General

Note: HVAC and BMCS contractors shall be responsible for coordinating inter-related works and no waver of responsibility is allowed. Advise the project officer any conflict between the scope and responsibility.

1. Utilise the existing points and field devices as necessary.

2. Advise any additional points and field devices necessary to form a complete operation of the entire system.
3. Extend the existing system and re-engineer and reprogram to completely control the new chilled water system.
4. Integrate the new Chiller BACnet High Level Interface gateway with the BMS to operate as specified and provide the Device Instance and IP details and shall carry out all necessary programming to ensure the current Chiller Staging strategy is maintained.
5. Test and commission the new installation.
6. The BMCS contractor shall allow progressive fine-tuning of control logic after Commissioning up to the end of the Defects Liability Period, without any additional cost to the Contract.

8. WATER TREATMENT

8.1 General

Water Treatment shall be to the Water Treatment Design Guide requirements. A dosing pot mounted on a plinth with SS Tray under the feet and drained to waste will be provided for each closed system. A set of corrosion coupons are to be provided for each closed system and tested each three months.

9. ELECTRICAL

9.1 Requirements

All equipment supplied and all work done shall be in accordance with this specification and the S.A.A. Wiring Rules and any other S.A.A. publications covering design, manufacture or installation of electrical equipment and to the requirements of the local Electricity Supply Authority. The latest issue or amendment of the relevant S.A.A. publication shall be used.

The switchboard manufacturer shall have a type test certificate in addition to Part Type Test Assembly certificate from switchgear supplier to which the design, manufacture and performance of the switchboard shall conform. The installation Contractor shall have an Electrical Contractors license and employ only licensed electricians on Site.

9.2 Switchboards

The S.C.A. shall comply in all respects with AS 61439.

Proof of performance of the board shall be substantiated by switchboard manufacturers' submission of a type test certificate from a qualified testing laboratory.

The section for switching of sub-system, circuitry shall be labelled 'Authorised Personnel Only', this section shall not allow any person operating switches and circuit breakers to come into contact with live parts.

All other sections of the SCA shall be labelled as indicated on switchboard drawings.

LED pilot light "Run" and "Fault" indication for all main components will be provided including supply and exhaust fans. Lamp test button is also required.

Individual pilot light 'fault' indication for all safety controls will be provided.

9.3 Connections

All cables unless enclosed shall be neatly arranged and loosely tied and supported neatly, where necessary.

All control cables to and from the switchboards shall terminate at an approved terminal block. No more than two cables shall terminate at any one terminal point. Power cables for Compressor and Condenser motors may be directly terminated on respective Contactors/over-loads.

The switchboard shall comprise a factory made and assembled, free standing floor mounted front connected enclosure with switchgear and control gear arranged as to industrial current standard.

On Site assembly of switchboard shall not be permitted.

The switchboards shall be totally wired to terminations as detailed above such that connections terminate at location and terminals designated and marked for the appropriate cables.

All wires and cables (unless otherwise specified or subject to statutory fire rating) shall be V75 thermoplastic insulated copper stranded of Australian manufacture.

Access for conduits and ducts shall be provided by removable aluminium gland plates.

Access to the front of the switchboard shall be by means of sheet metal doors fully dust sealed.

9.4 Field Wiring & Cabling Installation

All wiring in the mechanical services plant room and building areas will conform to the requirements of the Electrical Design Guide and AS3000.

Wiring shall be run in conduit, cable trunking and on trays or cable ladder to suit the application and all trays, trunking, cable ladder and troughing will be painted orange as per painting requirements for electrical items.

All external equipment will be connected using anaconda flexible steel sheathed conduit with appropriate fittings to produce a watertight connection. Flexible PVC conduit will not be used externally unless rated for that duty and will have a smooth external coating for protection such as Clipsal sheathed flexible conduit type CMS. No flexible conduit will be longer than 0.5 Metres long. This applies to controls connections as well.

Tray, ladder and cable supports must be suitable for the load applied and will conform to Australian Standard AS3000. Also project specific requirements for seismic restraints requirements and compliance.

All wiring penetrating a fire wall shall be installed using a special fire rated intumescent fitting.

10. TESTING & COMMISSIONING

10.1 General

Provide all equipment, labour, materials, power, consumables, etc. to conduct testing and commissioning. The Contractor shall carry out his own tests to verify operation and performance before requesting Council to attend inspections. Submit fully detailed Test Log sheets for inspection prior to any testing taking place. The Test Log sheets shall be A4 in size and present a clear and precise record of all tests to be carried out.

10.2 Testing

The following tests and checks shall be carried out by the Contractor:

1. Visual inspection of pipework and joints for leaks.
2. Water-test drains and overflow piping for leaks and for unobstructed drainage.
3. Check all power cabling terminations and megger test each circuit.
4. Verify integrity of all controls cabling to ensure electrical continuity and freedom from shorts and grounds.
5. Sequence checks of all specified control functions and cycles, such as Start, Stop, Chiller Staging, Cooling and Heating Cycles, etc.
6. Water flow rates shall be balanced to +5% / -0% of those scheduled. Measure and record Chilled Water flow rates, compare with scheduled design water flows, and submit.

Make all necessary adjustments and record all temperature set points and differentials on the Test Log sheets.

11. WARRANTY

Warranties for installation and equipment shall be for a period of 12 months minimum.

12. MAINTENANCE & SERVICE

The Installation Contractor is responsible for carrying out all preventative maintenance and servicing for the duration of the Defects Liability Period.

The mechanical contractor shall allow in the tender for maintenance and servicing from date of practical completion to the end of the maintenance and defects liability period, to comply with requirements of AS 3666, Local Authority and any other statutory body having jurisdiction.

Maintenance, in addition to rectification of faults, emergency service and carrying out of capacity tests as required, shall also include a minimum of 12 regular monthly visits for routine maintenance.

13. SCHEDULE OF PERFORMANCES

13.1 Circulation Water Pump

Type	Vertical Inline
Duty Flow	7.9 L/S
Duty Head	10 M
Power Supply, V/Ph/Hz	415/3/50
Sound Level @ 1m	49 Db (A)
Number of Pumps	2
Make / Equal	Baker Line

13.2 EC Fan

Fan no.	Description	Fan type	Volume L/s	Static pressure Pa	Phase /Voltage
EF 1-1	Exhaust air - AHU 1	EC centrifugal model plug fan	2750	600	3Ph/ 380-480V
EF 1-2	Exhaust air - AHU 1	EC centrifugal model plug fan	2750		
EF 1-3	Exhaust air - AHU 1	EC centrifugal model plug fan	2750	600	3Ph/ 380-480V
EF 1-4	Exhaust air - AHU 1	EC centrifugal model plug fan	2750		
EF 3-1	Exhaust air - AHU 3	EC centrifugal model plug fan	2750	600	3Ph/ 380-480V
EF 3-2	Exhaust air - AHU 3	EC centrifugal model plug fan	2750		
EF 3-3	Exhaust air - AHU 3	EC centrifugal model plug fan	2750	600	3Ph/ 380-480V
EF 3-4	Exhaust air - AHU 3	EC centrifugal model plug fan	2750		

13.3 AHU-1 and AHU-3 Coil Schedule (Identical)

	EVAPORATOR COIL-1	EVAPORATOR COIL-2	RUN AROUND COIL-1	RUN AROUND COIL-2	CONDENSER COIL-1
Duty	Cooling	Cooling	Heat Recovery	Heat Recovery	Heating
Nominal Total Capacity	84	84	120	120	275
Nominal Sensible Capacity	54	54	62	62	-
Nominal Air Flow (L/S)	5500	5500	5500	5500	11000
Ent Air DB (°C)	24.0	24.0	23.0	23.0	17.7
Ent Air WB (°C)	19.0	19.0	19.5	19.5	-
Ent Water Temp (°C)	6.0	6.0	6.0	6.0	45.0
Lvg Water Temp (°C)	-	-	-	-	-
Nominal Water Flow Rate (L/S)	2.3	2.3	7.5 @100kPa	7.5 @100kPa	13.4
Est. Water Pressure Drop (kPa)	21	21	11.5	11.5	24
Tubes Height	83	83	83	83	83
Rows Deep	6	6	3	3	5
Fins per Inch	12	12	9	9	11
Rows	6	6	3	3	5
Circuit	21	21	62	62	69
Est. Overall Dimensions in mm	500 x 2640	500 x 2640	1290 x 2550	1290 x 2640	1220 x 2640
Location	AHU-1 & 3	AHU-1 & 3	AHU-1 & 3	AHU-1 & 3	AHU-1 & 3
Number of Coils each AHU	1	1	1	1	1

Note: the coils data are for pricing purpose only and subject to final selection of the heat pump chiller and available space in the AHU plenums.